

80's → as a profession, Colombia

SYSTEMS ENGINEERING

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.

cavirguezs@udistrital.edu.co

• Computer
Sciences

• Software
Engineering

• Computer
Engineering

• Telecommunications
Engineering
!

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering

Universidad Distrital Francisco José de Caldas

2026-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System



Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System



What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance of complex systems**.
 - This **discipline** is based on interdisciplinary fields such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
 - **Systems Engineering** is a holistic approach to engineering that focuses on how to **design** and manage complex systems over their lifecycle.
- big
hard to read*



What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields such as **control engineering**, **industrial engineering**, **software engineering**, **mechanical engineering**, **electrical engineering**, **organizational studies**, **project management**, and **others**.
- **Systems Engineering** is a **holistic** approach to **engineering** that focuses on how to **design** and **manage** **complex systems** over their **lifecycle**.



What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
 - This **discipline** is based on interdisciplinary fields such as **control engineering**, **industrial engineering**, **software engineering**, **mechanical engineering**, **electrical engineering**, **organizational studies**, **project management**, and **others**.
 - **Systems Engineering** is a **holistic** approach to **engineering** that focuses on how to **design** and manage **complex systems** over their **lifecycle**.
- Handwritten notes:* "problem" with an arrow pointing to "holistic", and "all" with an arrow pointing to "complex systems".



Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for designing, implementing, and maintaining complex systems.
 - A Systems Engineer must have a broad understanding of engineering, mathematics, science, and technology.
 - A Systems Engineer must be able to analyze and solve problems in a systematic and logical manner.
- software engineering



Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for **designing**, **implementing**, and **maintaining** complex systems.
- A **Systems Engineer** must have a **broad understanding** of **engineering**, **mathematics**, **science**, and **technology**.

- A **Systems Engineer** must be able to analyze and solve problems in a systematic and holistic way.

representation

inspiration

Circuits
software
robotics



Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for **designing**, **implementing**, and **maintaining complex systems**.
- A **Systems Engineer** must have a **broad understanding** of **engineering**, **mathematics**, **science**, and **technology**.
- A **Systems Engineer** must be able to **analyze** and **solve problems** in a **systematic** and **holistic** way.

↓
P9-10-11



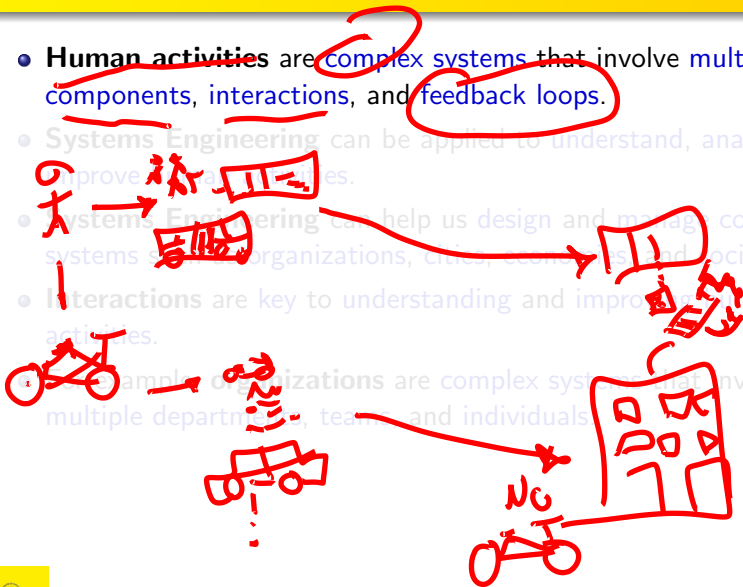
Outline

- 1 Basic Concepts
- 2 Systems Engineering**
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System



Human Activities

- **Human activities** are **complex systems** that involve **multiple components, interactions, and feedback loops**.
- Systems Engineering can be applied to understand, analyze, and improve **human activities**.
- Systems Engineering can help us design and manage complex systems such as organizations, cities, economies, and societies.
- **Interactions** are key to understanding and improving human activities.
- **Complex organizations** are complex systems that involve multiple departments, teams, and individuals.



Human Activities

- **Human activities** are complex systems that involve multiple components, interactions, and feedback loops.
- **Systems Engineering** can be applied to understand, analyze, and improve human activities.
- **Systems Engineering** can help us design and manage complex systems such as organizations, cities, economies, and societies.
- **Interactions** are key to understanding and improving human activities.
- For example, **organizations** are complex systems that involve multiple departments, teams, and individuals.

education system
health system
work system

!



Human Activities

- **Human activities** are complex systems that involve multiple components, interactions, and feedback loops.
- **Systems Engineering** can be applied to understand, analyze, and improve human activities.
- **Systems Engineering** can help us design and manage complex systems such as organizations, cities, economies, and societies.
- **Interactions** are key to understanding and improving human activities.
- For example, **organizations** are complex systems that involve multiple departments, teams, and individuals.



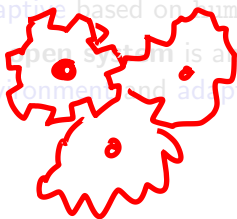
Organizations as Systems

- **Organizations** can be viewed as: a rational system, a natural system, or an open system.
- A rational system is a formal organization that is designed to achieve specific goals as a machine.
- A natural system is an informal organization that is emergent and adaptive based on human interactions.
- An open system is an organization that is interconnected with its environment and adapts to changes.



Organizations as Systems

- **Organizations** can be viewed as: a rational system, a natural system, or an open system.
- A **rational system** is a formal organization that is designed to achieve specific goals as a machine. *old - classical*
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
- An **open system** is an organization that is interconnected with its environment and adapts to changes.



Organizations as Systems

- **Organizations** can be viewed as: a rational system, a natural system, or an open system.
- A **rational system** is a formal organization that is designed to achieve specific goals as a machine.
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
- An open system is an organization that is interconnected with its environment and adapts to changes.

R.E.H.H. → Human Resources (rational)
X

Human Talent → rational (one-to-one)



Organizations as Systems

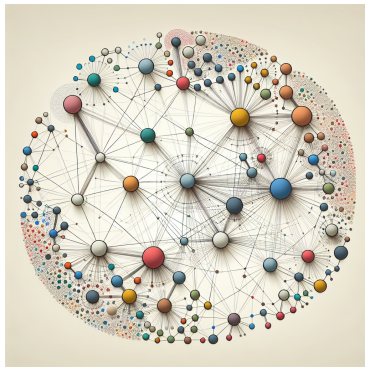
- **Organizations** can be viewed as: a rational system, a natural system, or an open system.
- A **rational system** is a formal organization that is designed to achieve specific goals as a machine.
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
- An **open system** is an organization that is interconnected with its environment and adapts to changes.

gas
electric ?



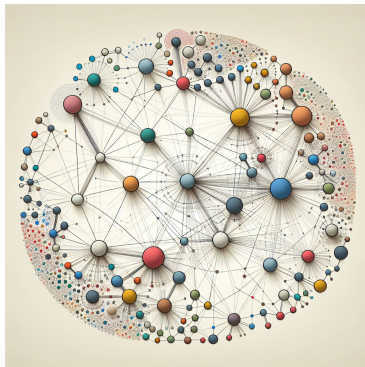
Synergy

- **Synergy** is a simple but powerful concept: **the whole is greater than the sum of its parts.**
- It means the **interactions** can **boost the capabilities** of the **parts** of the **system**. Also, it allows both understanding **emergent behaviors** and defining **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you can define a **computational machine**.



Synergy

- **Synergy** is a simple but powerful concept: **the whole is greater than the sum of its parts**.
- It means the **interactions** can **boost the capabilities** of the parts of the **system**. Also, it allows both understanding **emergent behaviors** and defining **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you can define a **computational machine**.



Real Madrid Galácticos: Systems Engineering Case Study

- **Galácticos Era** (2000 — 2006): Real Madrid assembled the most expensive and talented individual players in football history.
- **Star Players**: Zinedine Zidane, Robinho, David Beckham, Luís Figo, Ronaldo, Kaká, Roberto Carlos, ... each a world-class individual performer.
- **Expected Result**: With the best components, success should be guaranteed, right? Watch this video.
- **Real Result**: Only one Champions League in six years, two La Liga titles, but multiple early Champions League exits. Poor team chemistry and lack of system integration despite massive investment.
- **Systems Lesson**: A collection of excellent *parts* does not automatically create an excellent *system*. Synergy and integration matter more than individual talent.

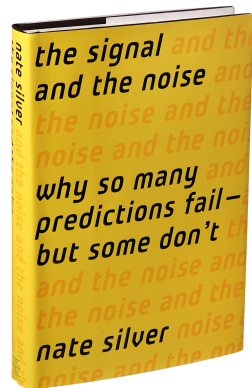
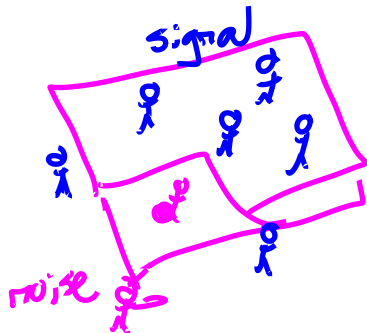


Real Madrid Galácticos: Systems Engineering Case Study

- **Galácticos Era** (2000 — 2006): Real Madrid assembled the most **expensive** and **talented** individual players in football history.
- **Star Players:** Zinedine Zidane, Robinho, David Beckham, Luís Figo, Ronaldo, Kaká. Roberto Carlos, ... each a **world-class** individual performer.
- **Expected Result:** With the best **components**, success should be **guaranteed**, right? Watch this video. *5 lost* *4 lost*
- **Real Result:** Only **one** Champions League in **six years** **two** La Liga titles, but multiple **early** Champions League exits. Poor **team chemistry** and lack of **system integration** despite massive investment.
- **Systems Lesson:** A collection of excellent **parts** does **not** automatically create an excellent **system**. **Synergy** and **integration** matter more than individual talent. *Interactions*



Synergy: Money Ball - Johan H. / Brad Pitt



Talking with Machines!

19²(12)-19(54)

Alonzo Churchill

Computer - Máquina

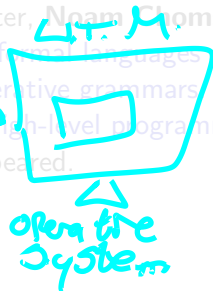
↳ Graphs ~ (1935)

Algo ↗

~ 1930 (Ph.D)

Alan Turing proposed a century ago a Universal Machine, capable of taking any algorithm defined as a state machine, and processing it in a binary language.

Forty years later, **Noam Chomsky** proposed the formal languages based on generative grammars. This is where the high-level programming languages appeared.



Talking with Machines!

lingüística

S + V + C

S → I / you / he / she / <substantive>
 <substantivo> → student / professor...

V →
 C →

- **Alan Turing** proposed a century ago a **Universal Machine**, capable of taking any **algorithm** defined as a **state machine**, and processing it in a **binary language**. 60-70s
- Forty years later, **Noam Chomsky** proposed the **formal languages** based on **generative grammars**. This is where the **high-level programming languages** appeared.

rules



Programming Languages

- **Programming Languages** with more capabilities and easier comprehension have been created. Also, more people started to code in specific domain programming languages.

Demo time!

simple
↓
low code
↓
LLM (codex)

- Andrej Karpathy, a leading figure at Tesla and now at OpenAI, said: *Nowadays, English is the most important programming language.*



Programming Languages

- **Programming Languages** with more **capabilities** and easier **comprehension** have been created. Also, more people started to **code** in specific **domain programming languages**.

Demo time!

- **Andrej Karpathy**, a leading figure at **Tesla** and now at **OpenAI**, said: *Nowadays, English is the most important programming language.*

vibe coding

→



Applications of Systems Engineering

- **Systems Engineering** can be applied to understand, analyze, and improve complex systems.
- **Systems Engineering** can be applied to design and manage complex systems such as organizations, cities, economies, and societies.
- **Systems Engineering** can be applied to design and manage complex systems such as software, hardware, and networks.

IoT



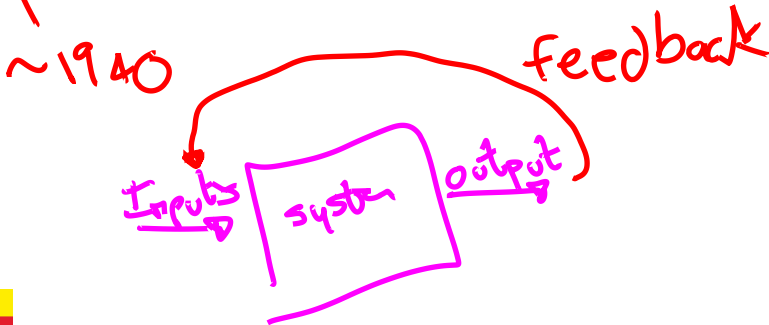
Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology**
- 4 Teams-Based Structure as a System



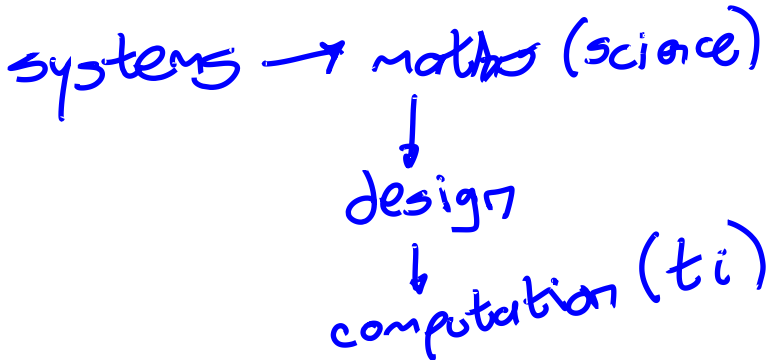
Cybernetics $\neq$ robotics

- **Cybernetics** is the study of systems, control, and communication in *animals, machines, and organizations*.
- **Cybernetics** is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.
- **Cybernetics** is the foundation of systems engineering and information technology.



Technology

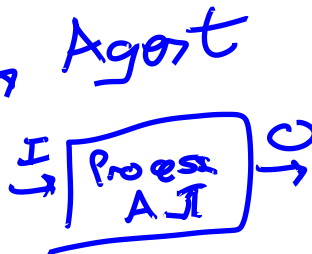
- **Technology** is the *application* of scientific knowledge to **solve** problems and **improve** systems. → **engineering**
- **Technology** is the key to **designing** and **managing** complex systems such as organizations, cities, economies, and societies.



Technology: AI

- **Artificial Intelligence** is a *field* of computer science that studies how to design and implement intelligent agents.

- Artificial Intelligence is the key to *designing* and *managing* complex systems such as smart-organizations, smart-cities, smart-economies, and smart-societies.



solve problems
like a human



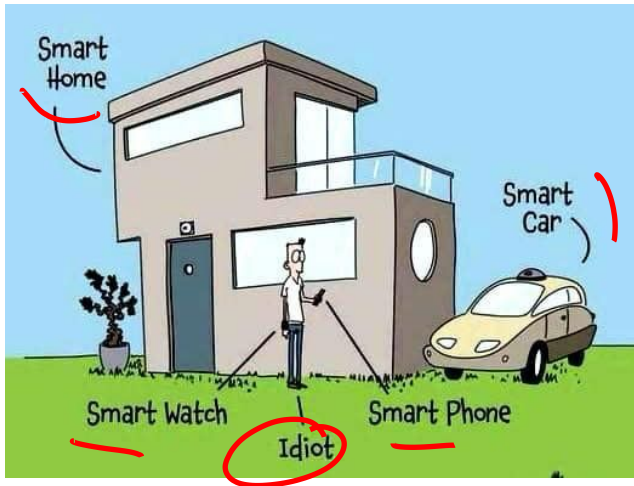
Technology: AI

- **Artificial Intelligence** is a *field* of computer science that *studies* how to *design* and *implement intelligent agents*.
- **Artificial Intelligence** is the key to *designing* and *managing* complex systems such as smart-organizations, smart-cities, smart-economies, and smart-societies.

all $\rightarrow$ a
system



Artificial Intelligence **not** as a System



Cybernetics & Technology in Systems Context

- **Cybernetics** and **technology** are the foundation of systems engineering and information technology.
- **Cybernetics** and **technology** are the foundation of artificial intelligence and smart systems.



Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System**



Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to **make decisions** and **solve problems**.
- **Team-based structure organizations** are flexible, agile, and innovative because they **empower employees** and **encourage collaboration**.



Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to **make decisions** and **solve problems**.
- **Team-based structure organizations** are flexible, agile, and innovative because they **empower employees** and **encourage collaboration**.



Teams as a System

- **Teams** are a **system** where collaboration (synergy) and communication are **key** to **success**.
- Defining and automating the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.



"Soft Skills" in Engineering

- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and harmoniously with other **people**.
- Typical Soft Skills:
 - **Communication** skills (verbal and written).
 - **Teamwork** and collaboration.
 - **Problem-solving** and critical thinking.
 - **Adaptability** and flexibility.
 - **Time management** and organization.
 - **Leadership** and management.
 - **Emotional intelligence**.
 - **Creativity** and innovation.
 - **Conflict resolution**.
 - **Networking** and relationship building.
 - **Customer service** and client management.



Computer Analyst — Technical Skills & Responsibilities

● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.



Computer Analyst — Technical Skills & Responsibilities

● Skills:

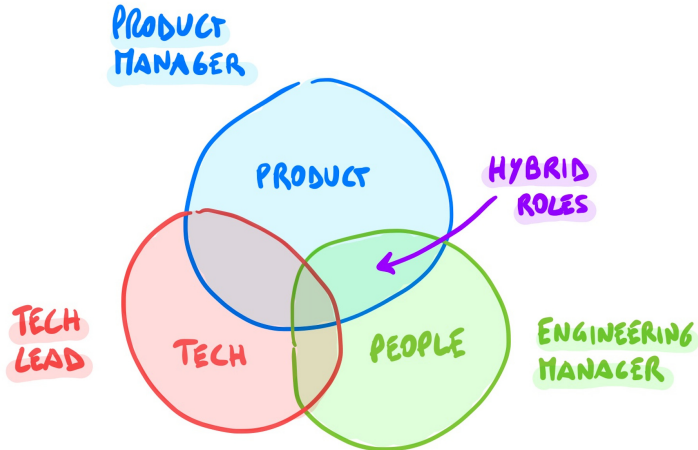
- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

● Responsibilities:

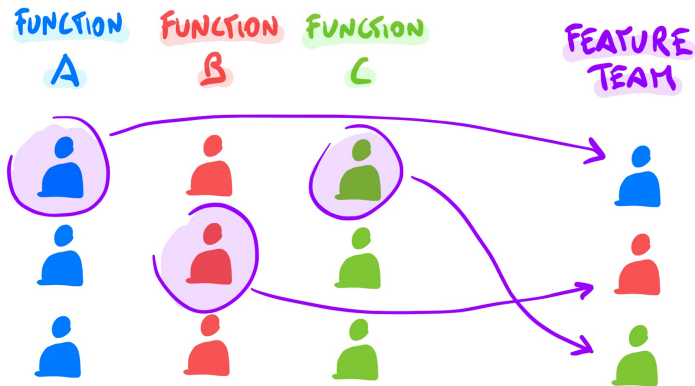
- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.



Technical Team Typical Structure



Feature Teams



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave people behind.



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave **people** behind.



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave people behind.



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to **fail**. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave people behind.



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to **fail**. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave people behind.



What does it mean to be a leader? II

- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for **self-growth** are **vital**.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



What does it mean to be a leader? II

- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are **vital**.
- Making **ethical decisions** is key; it leads to taking the **right and better actions**.



What does it mean to be a leader? II

- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are **vital**.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



What does it mean to be a leader? II

- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are **vital**.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



What does it mean to be a leader? II

- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are **vital**.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



What does it mean to be a leader? III

- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



What does it mean to be a leader? III

- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: Curiosity, Courage, and Commitment.

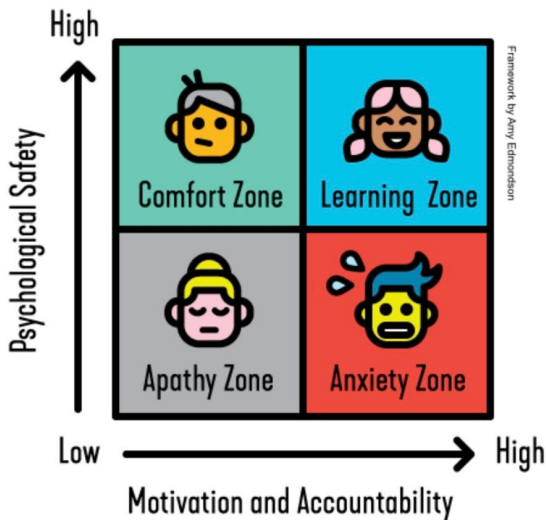


What does it mean to be a leader? III

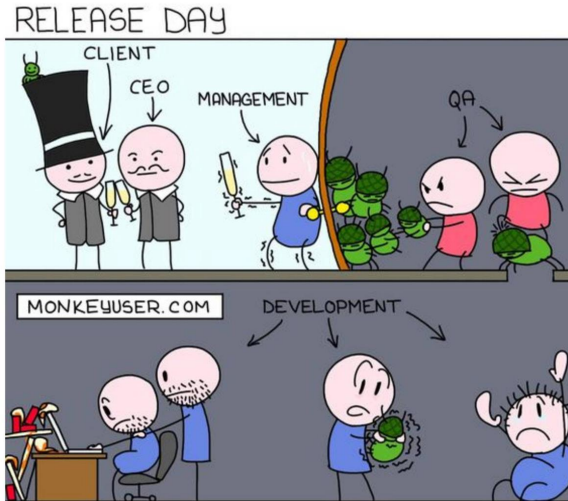
- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To **develop as a leader**, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



Working Zones



Real World!



Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

